

CS 204 – Advanced Programming – Spring 2026

3 credits

Prerequisite CS201. Although formally the grade D is prerequisite, those who got C+ or less are recommended to repeat CS201 and then take CS204, because the level of C+ or less in CS201 does not give enough background to succeed in CS204.

Description and objectives

This course aims to provide programming experience and to give advanced programming techniques. In this way, students would be more prepared to data structures and several other junior and senior level CS courses. CS204 is a prerequisite course for several CS courses including data structures. Thus, it is a must course for CS students and students who will take advanced CS courses.

CS204 heavily depends on CS201. Thus a very good CS201 background is needed. We will **NOT** make a review of CS201 topics.

The programming language that will be used in this course is C++; we will use CLion over Windows operating system as the development environment. You have to open a student account <https://www.jetbrains.com/clion/> using your sabanciuniv emails in order to activate free student license of CLion. More details about the development environment will be given later in this syllabus.

Topics planned to be covered

- ❑ Introduction (overview of basic concepts, C++ environment, preprocessor directives, compiler, compiler options, linker, libraries, debugging)
- ❑ Pointers and dynamic memory allocation
- ❑ Linked lists
- ❑ Stacks and queues
- ❑ Advanced issues on classes and object oriented programming
- ❑ Templates, templated classes and functions
- ❑ Data representation, bitwise operations
- ❑ Inheritance, polymorphism and advanced object oriented design
- ❑ Exception handling
- ❑ Programming with threads
- ❑ Visual programming and graphical user interfaces (if time and scheduling permits)

Instructor

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Office Hours: whenever I am in my office (generally I am in my office). Whenever you have a question you can drop by or write me.

Assistants: Detailed assistant information (office hours, etc.) will be available at SUCourse. All office hours are planned to be held online. See the details at SUCourse.

Textbook(s)

- Main texts** are "Ivor Horton's Beginning Visual C++ 2012", by Ivor Horton, ISBN: 978-1-118-36808-4.
Actually any version of this book is OK, I said 2012 since it can be found electronically. Although this book says "Visual C++", it is general enough.
"A Computer Science Tapestry" (CS201 book)
- Reference books** are "Starting out with C++ Early Objects", 7th edition, by T. Gaddis, J. Walters, G. Muganda
"Objects, Abstraction, Data Structures and Design using C++" by Koffman and Wolfgang.

We may not stick to the textbooks; you are responsible material covered in class too. Thus it is very important to attend to lectures. Moreover, most students do not need books in this course.

Schedule

Lectures: Monday 16:40-18:30 FENS G077 (auditorium), Wednesday 12:40-13:30 FMAN G071 (auditorium).

Labs: Sections A1, A2, A3: Thursday 12:40 – 14:30, B1, B2, B3: Friday 10:40 – 12:30,
C1, C2, C3: Friday 15:40 – 17:30.

All labs will be held physically and will start this week. Lab TAs are/will be listed at SUCourse. Some lab sections might be closed, stay tuned.

Homework Assignments

There will be 6 programming homework assignments. There will be no late submissions. You must submit your own work! No group works! No direct help from any online or AI source is allowed. All of these acts will be considered as plagiarism. Collaboration will not be considered as an excuse for plagiarism!

See below sections for homework grading and submission related issues.

Grading and Exams (absolutely tentative)

2 Midterm Exams (25% each): Midterm 1: April 6, 2026, Monday, 16:40 – 18:30.

Midterm 2: May 9, 2026, Saturday, 13:00 – 15:00.

Final (40%): will be scheduled by student resources.

Current plan is to have two Midterm exams and one Final exam. In case of changing plans to have a single Midterm Exams, total of exams (90%) will remain intact but the distribution will change.

No handheld electronic devices (phones, tablets, smart watches) are allowed in the exam rooms. They have to be left outside. Moreover, you are not allowed to wear any type of watches during the exam. Any violation of these rules will be considered as attempt to cheating.

Homework assignments (10%) – The homework assignments are of equal weight. All of them will be considered in grading. Although I plan to assign 6 homework assignments, the last one may not be graded; I will decide on the fate of the last homework towards the end of the semester. Homework grading will mostly be based on correctness of the execution; not based on how big is the mistake or how hard you work for it. Thus please test your code very carefully. No debugging will be done during grading and we will not try to understand where the problem is. No code/file changes are allowed after the submission. If you submit wrong file, we cannot grade your work. If you have mistyped even a single character that causes a compilation error, we cannot do anything. Even if you find out your mistake later, we do not allow code changes. See below sections for more about homework grading and submission.

Attendance (-2%): We will take attendance in both lectures and labs (starting after add-drop period). 70% attendance is required. Those who do not meet this threshold will be proportionally penalized up to 2%. No medical reports or other excuses will be considered as extra right for non-attendance; they are to be considered within 30% of right to not attend.

Use of SUCourse and Communications

We will make announcements via SUCourse that you will also receive as emails. Some announcements may be sent as plain email or the quick ones via WhatsApp group.

I will start a particular SUCourse announcement for you to ask general questions to the class and to me.

All lecture materials, homework and assignments will be posted at SUCourse. The submissions will also be there unless otherwise stated.

Lecture materials will be posted as PowerPoint file without annotations made in class. Sample codes discussed in class will also be shared. Weekly postings will be done after the lectures (Wednesdays or Thursdays).

Other Rules and Remarks

- The highest grade you can get from an **individual** homework assignment is 1.4 times of your weighted exam average. For example, if your weighted exam average (Midterms and Final) is 71.43% or more, then you may get up to 100 from any of the homework assignments. As another example, if your weighted exam average is 40%, then maximum of each homework is 56 even if you get more than that during grading. We will grade the homework assignments first without considering this, but at the end of the semester, after final exam is announced, we will apply the reductions.
- Students are expected to attend all classes, recitations/labs. In CS204 you will continue to learn a programming language and it is much easier to learn a programming language when a person explains it. You have to spend more time to compensate a missed class. So absenteeism does not buy time.
- Students are responsible for the material covered in class even if it is not part of the lecture notes published on the website (Sometimes we explain some stuff by writing). That is why attendance is important.
- Use of online sources (such as YouTube videos, code repositories, programming sites, ChatGPT, Gemini, Claude, etc.) is not recommended since they may lead you some materials/methods that are not taught (thus not allowed) and are not permitted. Moreover, using such resources may end up plagiarism in homework

assignments. An exception to this recommendation would be using online help to find the reason of an error message, but be careful about the issues mentioned above. Moreover, all recommendations that you find online must be applied with proper understanding. Solving a problem, writing a code without understanding the reasons and mechanisms behind them is nothing but pure self-deception and does not lead you to success.

- Students are responsible to check their emails (Sabanci University accounts) and SUCourse daily for any announcements related to this course.

Homework Submission and Grading Policy of CS 204

- All homework submissions must be done before the specified deadline. If you submit on the exact deadline time, then you are late. That means if the deadline is 21:00, then the latest second that you can submit is 20:59:59.
- Late homework submission is not allowed even with a penalty.
- Medical reports, or any other excuses (even valid ones) do **not** add extra time to the homework deadlines.
- Homework submission must be done only to SUCourse. Email or other type of submissions are not accepted.
- **No modification is allowed after SUCourse closes.** Thus, please make sure that you've submitted the correct homework. If you submit wrong file or if a crucial file is missing, we cannot grade your homework and you get zero (0). Sometimes, the students argue the local computer's system time as a proof of timely completion of a wrongly submitted homework; we cannot accept such local proofs and we cannot accept any file other the ones submitted to SUCourse. No exceptions to this rule.
- We do not allow code changes after SUCourse closes either. We grade whatever you submit to SUCourse. This rule holds even if one character change increases your grade from 0 to 100.
- **Homework grading is mostly based on correct execution.** We generate some test data for grading. If your homework executes wrongfully against this test data, your grade will be lowered accordingly no matter how big is your programming bug and no matter how hard you work for it. Sometimes the bug is too simple, but it may cause your program to work totally wrong and you may get very low grades. Sometimes, you mistype a character that causes some compilation errors. These are too unfortunate but we have nothing to do. In order to avoid such cases, please test your program thoroughly before the submission and make sure that the submitted version is the most up to date one. And please do not write us to complain about this policy and its outcomes within the semester; you know it at the beginning of the semester and you take this course by knowing this policy.
- **We do not debug your code during grading to understand where the problem is.** As mentioned in the previous bullet, we just test your code and grading is not based on how big your bug is.
- **Use of CodeRunner:** We plan to use CodeRunner embedded to SUCourse for most of the homework assignments for *functional grading* purposes (except the ones that require manual check such as multithreading applications). CodeRunner also allows you test the code before the final submission and you get feedback about the correctness against some test cases. There will be both *visible* and *hidden* test cases in CodeRunner, and both will contribute to the functional grade of your homework. You will get feedback for both visible and hidden test cases; but for the hidden test cases, you will not be able to see which parts are wrong. Moreover, we will definitely will not disclose details of the hidden test cases.
- **Other grading criteria** include modularity of the code, coding style, commenting, efficiency, etc. There might be homework-specific rules as well. Thus even you pass all test cases, your grade may be less than 100 if you do not follow stylistic and other rules.
- **Testing is your responsibility.** Depending on the suitability of homework, we will provide sample runs, together with the homework specification, corresponding to the visible CodeRunner test cases. However, we do not guarantee that sample runs / visible CodeRunner test cases cover all of the edge and corner cases. Also keep in mind that the hidden test cases will be different than the visible ones and we keep our right to add extra test cases after the submissions closes. To sum up, it is your responsibility to test your program against all possibilities in order not to lose points due to some bugs.
- **Only course material:** You have to use only the material covered in class. In other words, you are **not** allowed to use anything that has not been taught in CS201 and CS204. Thus, if you use any type of online help, you do not only take the risk of plagiarism, but also take the risk of violating this rule.
- **No break and continue:** You are **not** allowed to use `break` and `continue` statements.
- **No global:** You are not allowed to use global variables unless otherwise stated.
- Homework grading may be done via a demo during which you may be questioned about your homework.
- **Development platform related issues:**
 - o Formal development environment of CS204 is **CLion over Windows with G++ compiler and C++ 17 standard** (For technical help for installation, please refer to the guides at SUCourse). We had to pick this environment mainly due to two reasons: 1) CodeRunner uses G++, 2) University provides

Windows computers to professors and TAs. We, as the CS204 team, do not have a particular feeling for Mac users!

- **Attention Mac Users:** Default toolchain (compiler, etc.) that comes with MacOS is CLang and it might cause some compatibility problems due to undefined behaviors of C++. There is a workaround to install G++ on MacOS that we explain in the guides that we post at SUCourse. This is the only help that we can provide to Mac users since if it does not work, there is really nothing to do (in some Mac configurations this does not work). Thus, if this workaround does not work, the only option for you to continue with default Clang compiler.
- MS Visual Studio has its own compiler and may cause some inconsistencies with G++ of CLion.
- If you want to use MS Visual Studio, VSCode, Xcode (MacOS), CLion on MacOS with Clang compiler, or another platform, I have to say "use at your own risk". That means, we may not help in case of problems.
- It is not that easy to install any development environment on MacOS that uses G++. We provide some guides in SUCourse in order to help the Mac users to install CLion with G++ on MacOS, but we cannot provide any help during office hours if these guides do not work for your Mac.
- I know that Mac users prefer to use Xcode, but unfortunately Xcode is extremely problematic when it comes to CS204 topics. Xcode is forgiving for programmer mistakes and even if you think that your code works, there might be some logical errors in your code that Xcode suppresses. That is why, please do not use Xcode.
- However, even our formal platform (CLion over Windows) may cause some minor compatibility issues with CodeRunner due to differences in operating systems (CodeRunner runs on an Ubuntu 2204 server with G++ compiler). If you want to have zero compatibility issues, you need to use Ubuntu that we cannot assume to have for all students (thus cannot make it the formal environment).
- Those, who have problems such as working locally but not working over CodeRunner, can try onlinegdb.com for testing/debugging purposes.

Rules to submit a medical report to take a make-up exam in CS 204

- All medical reports must be issued or approved by the health center. Recently the rule of external reports has changed; private doctor reports are no longer accepted. Please check out health center for that. Please make sure that external medical reports are given to health center on the date of exam. If it is going to be delayed for some reason, please let the instructor know about it.
- If possible, let the instructor know about your medical issue and your unavailability for the exam before the exam. If there is a non-medical reason for missing an exam, this issue must be brought to instructor's attention before the exam.
- Those who will take the makeup exam should apply for it by following the course announcements; We will not follow up on the makeup exam takers via the medical reports. Please do not keep asking the date for the make-up exam and wait for my announcement about it.
- If you take regular midterm or final exam, even if you have a medical report for that day, the exam will be considered valid and makeup will not be given.
- There will be one makeup to cover one of the midterms, and another one to cover the final. You cannot miss both of the midterm exams. If you do so, makeup is given only for one of the midterms.
- There will not be a makeup exam to cover a makeup exam whatsoever.
- Last, but not the least, make-up exams will be **HARDER** than the normal exams for the sake of fairness to those who take the exam on time and studied less than the make-up takers.

CS204 AI Policy in a nutshell

- Course materials (lecture notes, assignments, sample questions, project spec., etc.) are provided only for your personal use. They must not be uploaded, shared, or given, wholly or partially, to any AI tool, service, or platform.
- Students may use AI tools only to reinforce their understanding of course topics (e.g., by asking clarifying questions or exploring related examples).
- Assignments, projects, and other graded work must be completed entirely by the student's own effort. Asking AI tools to directly or indirectly solve or generate answers for these tasks will be considered plagiarism.
- Any violation of these rules will be treated as an academic integrity issue and will be reported to the dean's office for disciplinary investigation.

Policy on Plagiarism, Homework Trading, Grading of AI generated Codes

A plagiarized homework may or may not be a verbatim copy of another homework. Verbatim copies are of course plagiarized ones. If you use any online tool or code provided by online sources, this is also plagiarism. If a homework is derived from another one by partially changing some parts, this action is also plagiarism. Cooperation and working together is not an excuse. A common fallacy is that the graders cannot catch a program that is developed by partially changing another program. Believe it or not, such programs are caught very easily. There are software products available to catch such cases, and we use them. When a plagiarism case is detected, sanctions are applied to all parties regardless of the actual source of the homework. This is simply getting 0.

AI generated homework assignments are also considered as plagiarized ones. AI generated codes really look alike and the plagiarism tools flag them. If you get flagged that way, your homework will get 0. No appeals will be accepted; if you do not want to take the risk of being flagged, then do not use AI for homework assignments.

If you get 0 two times, then all of the homework grades will be set to 0.

**Use of AI tools will also be considered as plagiarism for any grading item.
Plagiarized assignments will not be graded.**

For the exams, any type of cheating (including having an electronic device other than calculator in proximity) will be reported to the Dean's Office for disciplinary actions